

Lab 3: Compensator Design

Overview

In this lab, you will get the chance to do what you couldn't do last time around with proportional control and design a first-order compensator for your kittedicopter that satisfies stricter specifications.

Compensator Design

The specifications for your controller are as follows:

- Set point tracking to within 0.4m of the final value
- Overshoot <20%
- Settling time to 2% of final value < t_e , where t_e is a personalised (based on system) expected settling time given by the simulation program under "options"
- Settling of output disturbance rejections back to 2% of the initial (pre-disturbance) value < t_e
- Sinusoidal tracking to within 0.4m, with a 0.1 rad/s sinusoid.

You may also add specifications that you feel would add to the quality of the design or should be considered in the given scenario.

Your controller circuits from lab 2 will be returned to you, as they can easily be adapted to form first-order compensators. Instructions for this are provided in the circuit appendix released with these instructions.

Preparation

Collect your Veroboard controllers from Mr De Maar in the control lab and check what your assigned t_e value is, as these are dependent on the speed of your plant.

The bulk of this lab's workload can be done prior to your lab slot by designing your compensator and verifying the performance of your compensated plant in simulation.

Ensure your controller is designed and built **on Veroboard** before you arrive at the lab to test it.

Instructions for the lab session

The aim of this session is to just to evaluate the performance of your controller. As with lab 2, you will demonstrate your controller to Mr De Maar, who will subject your controller to various tests as per the specifications given.

Report

Use the provided template to write a brief report on your findings.

EEE3094S Lab 3 Report

The same general report-writing advice from the previous lab applies here. Please try to present your work clearly and professionally. If you wouldn't have a good time reading your report, the person marking it definitely won't. I remind you that your report shouldn't exceed **15 pages** (not including Appendices).

Controller Design

This section should include:

- A brief description of the system model you derived previously for context
- An explanation of why you have selected the specific type of controller you are going for.
- A description of your controller design process, supported by calculations and figures.
- Tests of your controller in simulation.
- Details of the physical design of your controller, e.g. a schematic, and photos of your final board.

Controller Tests

This section should include:

- A description and plots of the tests you performed to evaluate your controller experimentally
- An evaluation of your controller's performance against the given specifications, culminating in an ultimate conclusion about whether your controller worked as intended.
- Recommendations of how to improve the performance of your system (if applicable).

Add the results of your controller demo tests to the appendix of your report.

You may adapt the LaTeX template provided to you in Lab 2 for this report.

The rubrics for both the report and the controller demo are given below.

Report Rubric

Category	Fail [0-10]	Passable [10-12]	Good [12-15]	Great [15-20]
Controller Design [25]	Analysis/ the design process is totally incorrect, not based on anything learned in class or absent/ impossible to follow in the report. There is little to no engagement with the specifications. The design is not supported by simulations/ the simulations are performed poorly. MATLAB or similar software is used incorrectly, OR in a way that appears somewhat correct, but the outputs are not validated or are weakly validated. No/ very little consideration is given to the robustness of the design.	The analysis/ design process applies some concepts learned in class but they are not well-understood, and there are fundamental mistakes in the approach. The design is linked to the specifications (i.e. they are converted into pole placements/ gain values) but it is either incorrect OR there is no/ a weak attempt to assess the plausibility of the specifications. The design is not supported by simulations/ the simulations are performed poorly. MATLAB or similar software is used in a way that appears mostly correct, but the outputs are not validated or are weakly validated. No/ very little consideration is given to the robustness of the design.	The analysis/ design process is logical and well-grounded in the theory of the course. There may be some minor errors e.g. calculation mistakes, or small misunderstandings. Analysis is supported by some evidence. (Figures may have some minor flaws but are still readable.) The design is strongly linked to the specifications and the plausibility of the specifications is considered. There may be some small problems with this analysis or how it is documented. The design is supported by simulations. There may be some small problems with how these are executed or documented. If MATLAB or similar software is used, the outputs are still validated with manual calculations and correct explanations. Limited consideration is given to the robustness of the design.	The analysis/ design process is logical and well-grounded in the theory of the course, demonstrating a strong understanding, and calculations are accurately done. Analysis is supported by strong evidence that is well-documented. If MATLAB or similar software is used, the outputs are still validated with manual calculations and correct explanations. The design is strongly linked to the specifications, and the plausibility of the specifications is considered. This analysis is clearly explained and well-documented. The design is supported by simulations. These are executed sensibly and documented effectively. Consideration is given to the robustness of the design.
Controller Test [25]	As per demo rubric below.			

Controller Evaluation [30]	Results are not analyzed/ analysis is largely incorrect or incoherent. There are no figures or the figures are unintelligible/ not referred to in text. The performance of the controller is not evaluated or the evaluation is mostly incorrect. The original specifications are not referred to. No/inadequate comparison to proportional controller. No recommendations were made where they should have been, OR the recommendations made were not based on any evaluation of the controller's performance.	Analysis of the results goes in the right direction but is weak or with major flaws in understanding OR is not supported by figures/ figures are not clear. Conclusions about the performance of the controller are basically correct, but incomplete or not well-grounded in the results. There is no attempt or a weak attempt to link the evaluation to the specifications. There was an attempt to make recommendations based on the conclusions, but they are impractical, OR do not address the specifications and limitations of the control design.	Results were analyzed in a way that displays some understanding of the course material and a link to the results, but there may be some minor flaws in the interpretation OR in the way that the data is presented OR the figures are present and readable but not tied to the analysis. Conclusions about the performance of the controllers are mostly logical and well-supported by the results, and link back to the original specifications, but there may be some minor errors or things left out. Practical recommendations were made based on the conclusions drawn above, where applicable, but there may be some minor errors or things left out, OR they partially link back to the specifications and/or do not fully address the limitations in the control design	Results were analyzed logically, showing understanding of the course material and analysis was linked to clear figures. Conclusions about the performance of the controller are logical and well-supported by the results in the paper. The evaluation clearly links the results back to the specifications of the project, and the limitations of the controller design. Recommendations were made based on the conclusions drawn above, where applicable and are well-supported by the results of the paper. The recommendations are practical and seek to improve the controller's performance based on the specifications of the project and/or address the limitations present in the control design.
Report [20]	Report is unintelligible, very unprofessional or has a lot of missing information.	Report has all the basic information but there are some spelling, grammar or formatting issues.	Report presents all the basic information clearly and professionally. The key steps of the method are described, but some details may have been left out.	Report is clear and professional. Methods are described in sufficient detail to be fully repeatable. Decisions are well-motivated, with clear links drawn between text and figures.

EEE3094S Improved Controller Demonstration 2024

Student Name		Student Number	
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Controller Demonstration Marking Guidelines (Total Marks 25)						
Marks	0	1	2	3	4	MARK
Set Point Tracking	No Tracking or Large Error	Within 1m of Final Value	Within 0.8m of Final Value	Within 0.6m of Final Value	Within 0.4m of Final Value	
Over/undershoot	Unlimited/ Very Large	Within 30%	Within 25%	Within 20%		
Settling Time to 2% No oscillations	$> t_e + 7s$	$> t_e + 4s$ $< t_e + 7s$	$> t_e + 2s$ $< t_e + 4s$	$> t_e$ $< t_e + 2s$	$< t_e$	
Output Disturbance rejection. Return to the previous final value	$> t_e + 7s$	$> t_e + 4s$ $< t_e + 7s$	$> t_e + 2s$ $< t_e + 4s$	$> t_e$ $< t_e + 2s$	$< t_e$	
Sinusoidal input tracking at 0.1 Rad/sec	No Tracking or Large Error	Within 1m of Final Value	Within 0.8m of Final Value	Within 0.6m of Final Value	Within 0.4m of Final Value	
Simulated Plot for an input Step of 5.	No Plot		Plot shown	Plot shown. Meeting the specifications		
Circuit diagram of your controller	No circuit diagram	Partial circuit diagram	Circuit diagram not fully labelled	Fully labelled circuit diagram		
Penalty Marks	Controller Output: High amplitude oscillations of $> 0.4v$ 2 Mark penalty	On/Off Bang-Bang Controller Type 8 Mark penalty	Using more than 4-Op Amps 3 Mark penalty	Other: Mark capped at 50% if implemented on breadboard		
TOTAL MARKS						

Where t_e is expected settling time as shown under “options” in the simulation program.

Assessor Name:

Signature:

Date: